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ABSTRACT

Current music generative models can generate high-quality music, but does this ability imply that they “understand” how good the music is, and is that understanding aligned with human evaluation? Previous attempts to use the likelihood of a generative model to evaluate music, an approach commonly used in text, have proven unsuccessful, leading researchers to rely on standalone supervised music evaluation models. In this paper, we answer this question affirmatively: we show that a models intrinsic signals—derived from its hidden representations and predictions—are strongly correlated with human ratings. In particular, we study MusicGen and consider three types of features: (1) prediction loss, (2) prediction entropy, and (3) concepts extracted from the model using sparse autoencoders (SAE). Using these features, we train a lightweight prediction model to estimate subjective ratings. We evaluate these features both individually and in combination. Our underlying assumption is that these signals partially capture the listening experience: prediction loss and entropy reflect expectation, anticipation, and surprise, while SAE latent features capture higher-level musical representations that align with human judgments of good versus bad music clips. Experiments on five human-evaluation benchmarks—MusicEval, SongEval, MusicPref, AIME, and Music Arena—spanning continuous ratings and pairwise preferences, show that each intrinsic signal is individually predictive of human ratings, and that predictive and representational signals capture partially complementary aspects of human judgment—combining them helps on some benchmarks, while individual signals are already competitive on others.

1. INTRODUCTION

Generative music models have advanced rapidly, yet judging the musicality of their outputs without human listeners remains a stubbornly open problem. Recent years have seen a growing body of human-evaluation datasets for music generation—including MusicEval [1], SongEval [2], MusicPref [3], AIME [4], and Music Arena [5]—and of audio-based predictors that attempt to approximate these ratings, most notably Meta Audiobox Aesthetics [6] and

music-instruction benchmarks such as CMI-Bench [7] and its reward-model variant [8]. These approaches share a common assumption: musicality is assessed from the generated audio alone, treating the music model as a black box.

A second, less explored line of evidence points to the generative model’s internal signals. Token-level loss correlates only loosely with perceived music quality [9], and perplexity alone has been shown to be an unreliable indicator of prediction correctness [10]—a confidently-wrong step can look much like a cautiously-right one. Two further intrinsic signals from the same pretrained model—predictive entropy from the logits, and sparse-autoencoder (SAE) latents from the hidden states—expose aspects of the model’s state that loss alone does not. Predictive entropy describes how the model spreads its probability mass across candidate next tokens—the shape of its bet on what comes next, in the sense of Huron’s probabilistic account of musical anticipation [11]. Loss reports whether the prediction was accurate—low when the model assigned high probability to the ground-truth token, high when it did not. Entropy, in contrast, quantifies the uncertainty of the predictive distribution itself, independently of the target. Jointly, they separate confidently correct from confidently incorrect predictions. SAE latents, in turn, have been shown on a large music model to carry semantically interpretable, steerable concepts—each latent admits a musically meaningful descriptor [12]. That work uses such latents for control; we repurpose the same readout for evaluation, asking whether latents already named concept-by-concept also align with how humans rate generated music.

Our framing. We take a model-centric evaluation view. Rather than training an audio predictor on top of generated waveforms, we ask whether a generative music model, through its own internal states, already knows what humans will judge as good or bad music. We study three intrinsic signals from a pretrained music model: two logit-level predictive signals—per-step loss and next-token entropy—and one hidden-state-level representational signal—SAE latent features. Human evaluation scores, drawn from five datasets spanning continuous ratings and pairwise preferences, serve as the ground truth against which these signals are probed.

Contributions.

- **Intrinsic-signal alignment framework.** We show that model-internal signals can be directly aligned with human evaluation targets across five heterogeneous datasets under one unified protocol, with-



out requiring an audio-content encoder as the main front-end.

- **Mechanism-level finding: loss-entropy disagreement.**(TODO: need experiments support) We identify an *overconfident-wrong* regime (low entropy, high loss) that is the dominant source of human-negative ratings, and show that the disagreement between loss and entropy is itself a more informative feature than either alone.
- **Representational evidence from SAE.** We find that a subset of SAE latents, learned without any human-rating supervision, already encodes a good-versus-bad axis that can be read out linearly, yielding gains complementary to loss and entropy.

2. BACKGROUND AND RELATED WORK

We position our work along three axes: audio-based predictors and benchmarks for music generation quality, human-evaluation datasets that supply ground-truth ratings, and model-internal signals—loss, entropy, and sparse representations—that have been studied in language modeling and interpretability but rarely brought together for music evaluation.

2.1 Audio-Based Music Quality Predictors and Benchmarks

A first family of methods predicts human-perceived quality directly from the generated audio. Meta Audiobox Aesthetics [6] jointly models speech, music, and sound along four human-aligned dimensions (content enjoyment, content usefulness, production complexity, production quality), and has become a de facto audio-side baseline for generated music. CMI-Bench [7] broadens this axis to music *instruction-following*, providing a comprehensive benchmark that grades how closely generated music matches compositional specifications. More recently, CMI-RewardBench [8] evaluates music reward models under compositional multimodal instructions, moving closer to preference modeling but still operating on audio inputs. All of these approaches take the generated waveform as the primary input and treat the generative model as a black box, leaving its internal state unused.

2.2 Human Evaluation Datasets for Music Generation

Recent datasets differ substantially in their supervision semantics and evaluation dimensions. MusicEval [1] provides expert continuous ratings for text-to-music generations; SongEval [2] extends the continuous-rating paradigm to full songs with multiple aesthetic dimensions. In contrast, preference-based datasets collect pairwise judgments: MusicPref [3] aligns text-to-music evaluation with listener preferences, AIME [4] benchmarks a range of generation models and metrics through controlled human preference studies, and Music Arena [5] provides a live crowd-sourced evaluation platform analogous to Chatbot Arena. These datasets are individually valuable but use

heterogeneous supervision formats (absolute MOS, multi-dimension ratings, pairwise wins/losses), making cross-dataset comparison a non-trivial design question that we address in Section 3.

2.3 Model-Internal Signals: Loss, Entropy, and Calibration

A second family of evidence—central to our approach—treats intrinsic quantities of the generative model itself as diagnostic signals. Recent work [9] studies token-level loss in music large language models and reports a non-monotone relationship with perceived quality: local loss minima often do not coincide with listener-preferred regions, and injected noise can even lower the loss without improving quality. A complementary perspective from language modeling shows that perplexity (and therefore negative log-likelihood) cannot by itself distinguish confidently-correct from confidently-incorrect predictions [10]. This aligns with a long-standing observation in the calibration literature: modern neural networks are systematically overconfident [13], and penalizing overconfident output distributions acts as an implicit regularizer [14]. Entropy thus enters naturally as a complementary signal. It has been used as a reliability indicator at decoding time [15], and recent work on reinforcement learning for reasoning models establishes an empirical entropy-performance law $R = -a \exp(H) + b$ [16], confirming that entropy and output musicality are structurally linked rather than merely correlated. Finally, the teacher-forcing regime used to compute token loss is known to differ systematically from free-running inference [17], which partially explains why loss alone is an imperfect proxy for generation quality. Taken together, these strands suggest that loss and entropy carry related but distinct information about model confidence and correctness, and that their joint analysis is more informative than either signal in isolation.

2.4 Sparse Autoencoders for Representational Probing

Sparse autoencoders (SAEs) applied to hidden activations have emerged as an interpretability tool for identifying monosemantic latent directions in pretrained neural networks, with prior work in language modeling and vision using SAE latents to uncover features that correspond to human-interpretable concepts. Closest to our setting, [12] trains SAEs on a large generative music model and reports that the resulting latents admit semantic interpretation: each latent can be given a concise musical descriptor, and applying the latents as steering vectors at inference time produces audible—if modest—changes in the generated audio. Their focus is *controllability* through representational edits. We take the same representational readout but repurpose it for *evaluation*: instead of intervening on the model, we ask whether the latents that Singh et al. interpret concept-by-concept also organize clips along a good-versus-bad axis that correlates with human ratings. Relative to loss and entropy—both of which summarize

Five databases → preprocessing → train/val/test splits → {loss curve, entropy curve, SAE features} → {single models ×3, hybrid models ×4} → predicted quality scores → correlation with human ratings.

Figure 1. Simplified end-to-end methodology flowchart.

the output distribution—SAE features operate on the representational bottleneck and therefore offer a qualitatively different probing surface for whether the model *internally* represents properties that correlate with human judgment. We exploit this distinction in Section 5.3.

3. METHODOLOGY

We design a model-aware evaluation pipeline from five human-evaluation databases to human-aligned prediction targets. The pipeline includes dataset-specific preprocessing, split construction, three feature extraction branches (loss, entropy, SAE), and model training in both single and hybrid settings.

3.1 Data Processing Across Five Databases

We use MusicPref, AIME, MusicEval, SongEval, and MusicArena. MusicEval and SongEval provide continuous ratings (including multi-dimension or multi-annotator settings), while MusicPref, AIME, and MusicArena provide preference-style supervision. We keep each dataset’s supervision semantics during preprocessing, run deterministic train/val/test splits with fixed seeds, and support both clean and noisy tracks for robustness.

For all pairwise labels, we use one unified objective. For a pair (a, b) , s_a and s_b are latent scalar quality scores, and m is the ranking margin in the hinge term. Win and loss use symmetric ReLU penalties, while tie is optimized by minimizing score distance:

$$\begin{aligned}\mathcal{L}_{\text{win}} &= \max(0, m + s_b - s_a), \\ \mathcal{L}_{\text{lose}} &= \max(0, m + s_a - s_b), \\ \mathcal{L}_{\text{tie}} &= (s_a - s_b)^2.\end{aligned}\tag{1}$$

After optimization, each latent score s_i is mapped to a common 1–5 range. Here ϵ is a small constant for numerical stability:

$$\tilde{s}_i = 1 + 4 \cdot \frac{s_i - \min_j s_j}{\max_j s_j - \min_j s_j + \epsilon}.\tag{2}$$

For robustness experiments, noisy labels are generated by bounded Gaussian perturbation, where ϵ_i is zero-mean Gaussian noise with variance σ^2 :

$$y_i^{\text{noisy}} = \text{clip}(y_i + \epsilon_i, 1, 5), \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2).\tag{3}$$

3.2 Feature Extraction

We extract three complementary feature families.

Loss curve. Given a token sequence $\{x_t\}_{t=1}^T$, where t is the token-time index and T is sequence length, teacher-forcing token loss is

$$\begin{aligned}\ell_t &= -\log p_\theta(x_t | x_{<t}), \\ \mathbf{l} &= [\ell_1, \ell_2, \dots, \ell_T].\end{aligned}\tag{4}$$

where $x_{<t}$ is left context and $p_\theta(\cdot | x_{<t})$ is the model distribution parameterized by θ . $\mathbf{l}$ denotes the full loss curve across time. Besides the raw sequence, we use statistics such as mean, variance, segment-wise averages, and local peaks.

Entropy curve from logits. At each step, entropy of the next-token distribution is

$$\begin{aligned}H_t &= -\sum_{k=1}^K p_\theta(k | x_{<t}) \log p_\theta(k | x_{<t}), \\ \mathbf{h} &= [H_1, H_2, \dots, H_T].\end{aligned}\tag{5}$$

where K is codebook vocabulary size and $\mathbf{h}$ is the temporal entropy curve. To characterize uncertainty-musicality interactions, we also compute clip-level coupling between loss and entropy:

$$\rho_{\ell, H}^{(i)} = \text{corr}(\mathbf{l}^{(i)}, \mathbf{h}^{(i)}).\tag{6}$$

Here $\rho_{\ell, H}^{(i)}$ is the Pearson correlation within clip i .

SAE features. Let $u_t \in \mathbb{R}^d$ be hidden states at time t . SAE produces sparse codes $z_t = f_{\text{enc}}(u_t)$ and reconstructions $\hat{u}_t = f_{\text{dec}}(z_t)$ with objective

$$\mathcal{L}_{\text{SAE}} = \frac{1}{T} \sum_{t=1}^T \|u_t - \hat{u}_t\|_2^2 + \lambda \frac{1}{T} \sum_{t=1}^T \|z_t\|_1.\tag{7}$$

The first term is reconstruction error, and λ controls sparsity strength via the ℓ_1 penalty on z_t . From $\{z_t\}$, we build sequence-level and pooled descriptors for downstream prediction.

3.3 Model Training: Three Single Models and Four Hybrid Models

We define seven model families. The three single models are: *Single-Loss*, *Single-Entropy*, and *Single-SAE*. The four hybrid models are: *Loss+Entropy*, *Loss+SAE*, *Entropy+SAE*, and *Loss+Entropy+SAE*. Fusion is implemented by late concatenation or multi-stream temporal encoding. In late fusion, r_{loss} , r_{ent} , and r_{sae} are branch-level representations and the final feature is

$$r = [r_{\text{loss}}; r_{\text{ent}}; r_{\text{sae}}].\tag{8}$$

In architecture terms, we use a codebook-aware temporal representation. For each time step t , signals from four codebooks are kept separate and concatenated, rather than averaged at input time. With codebook-level feature $f_t^{(c)}$ for codebook index $c \in \{1, 2, 3, 4\}$ and validity mask $m_t \in \{0, 1\}$, the model input is

$$v_t = [f_t^{(1)}; f_t^{(2)}; f_t^{(3)}; f_t^{(4)}; m_t].\tag{9}$$

For hybrid settings, $f_t^{(c)}$ can include loss, entropy, or SAE-
projected channels depending on the branch, while m_t
consistently indicates valid (non-padding) positions.

The mask is applied both during feature preparation and
temporal pooling so that variable-length sequences do not
bias the representation. Here $\phi(\cdot)$ is the temporal encoder
and ϵ avoids division by zero:

$$\tilde{v}_t = m_t v_t, \quad r = \frac{\sum_{t=1}^T m_t \phi(\tilde{v}_t)}{\sum_{t=1}^T m_t + \epsilon}. \quad (10)$$

All predictors are trained as regression to human-
aligned targets $y_i \in [1, 5]$, where $\hat{y}_i$ is model output for
sample i and N is sample count:

$$\mathcal{L}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2, \quad (11)$$

and optional regularization terms when needed.

3.4 Evaluation Protocol

We evaluate predicted scores against human ratings using
Pearson correlation, Spearman correlation, and MSE on
held-out sets. In Pearson, $\hat{y}$ and $\bar{y}$ are sample means of
predictions and targets:

$$r = \frac{\sum_i (\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}, \quad (12)$$

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}. \quad (13)$$

In Spearman, d_i is the rank difference for sample i and n
is the number of evaluated samples. All single and hybrid
models are compared under identical splits and preprocess-
ing to ensure fairness.

4. RESULTS

We report a unified comparison between large-scale base-
lines and our internal-signal models under the same
evaluation protocol. The final numeric results (Pear-
son/Spearman on clean and noisy tracks) will be filled after
final reruns with fixed seeds and finalized checkpoints.

For aesthetics baselines (CE/CU/PC/PQ), we apply a
score-alignment step before comparison with human MOS.
Let z_i be the raw aesthetics output for sample i . We use an
affine mapping to the 1–5 target range,

$$\hat{z}_i = \text{clip}(a z_i + b, 1, 5), \quad (14)$$

where a and b are estimated on the validation split only,
then applied to test predictions. This keeps baseline and
proposed models on the same target scale while avoiding
test-set leakage in calibration.

5. ANALYSIS

While Section 4 reports aggregate predictive performance,
the three intrinsic signals differ in what they reveal about
the model’s internal state. We analyze them from three
complementary angles: a *calibration view* on when loss
and entropy disagree, a *temporal view* on how uncertainty
evolves during generation, and a *representational view* that
probes SAE latents for an interpretable good-versus-bad
axis.

5.1 Calibration View: When Loss and Entropy Disagree

Loss and entropy are both derived from the same output
distribution $p_\theta(\cdot \mid x_{<t})$ and are often treated as two faces
of the same coin. In a well-calibrated model they co-vary:
high-confidence predictions (low entropy) are usually also
correct (low loss). Modern neural networks, however, are
known to be miscalibrated and systematically overconfi-
dent [13], and penalizing confident outputs has been shown
to act as an implicit regularizer against this tendency [14].

We frame each generation step as a wager, follow-
ing Huron’s probabilistic account of musical anticipa-
tion [11]. At step t the model distributes a unit of prob-
ability mass across candidate next tokens; entropy $H_t =$
 $-\sum_k p_\theta(k \mid x_{<t}) \log p_\theta(k \mid x_{<t})$ summarizes *how* that
stake is placed—concentrated on a few candidates (low
 H_t) or spread across many (high H_t)—while token loss
 $\ell_t = -\log p_\theta(x_t \mid x_{<t})$ records *whether* the stake hap-
pened to land on the ground-truth token. Loss and entropy
therefore read out orthogonal aspects of the same bet, and
their four joint regimes correspond to the four ways a mu-
sical wager can resolve (TODO: full experimental support
for this taxonomy is still pending; the classification below
is predicted from the framing and sampled from prelimi-
nary runs, not yet validated end-to-end).

We observe the same pattern in generative music mod-
els. Partitioning all token steps into a 2×2 grid of {low,
high} loss $\times$ {low, high} entropy gives four regimes with
distinct human-rating signatures:

- **Low loss, low entropy** (confident correct): the ideal
regime, positively associated with high human rat-
ings.
- **High loss, high entropy** (uncertain wrong): the
model knows it does not know; often co-located with
genuine musical surprise such as section boundaries,
and does not strongly penalize ratings.
- **Low loss, high entropy** (surprise survived): diffuse
predictive mass where the ground-truth token never-
theless receives enough probability.
- **High loss, low entropy** (overconfident wrong): the
mechanism-level failure mode in which the model
commits to a sharply peaked prediction inconsis-
tent with the ground-truth token. This regime is the
strongest negative predictor of human rating among
the four.

Group	Model	Input Signals	P (Clean)	S (Clean)	P (Noisy)	S (Noisy)
Baseline	Aesthetics-CE	Audio signal (five databases)	TBD	TBD	TBD	TBD
Baseline	Aesthetics-CU	Audio signal (five databases)	TBD	TBD	TBD	TBD
Baseline	Aesthetics-PC	Audio signal (five databases)	TBD	TBD	TBD	TBD
Baseline	Aesthetics-PQ	Audio signal (five databases)	TBD	TBD	TBD	TBD
Baseline	Mean Loss	Mean token loss	TBD	TBD	TBD	TBD
Baseline	Loss Curve CNN	Loss curve + mask	TBD	TBD	TBD	TBD
Single	Single-Loss	Loss (4 codebooks) + mask	TBD	TBD	TBD	TBD
Single	Single-Entropy	Entropy (4 codebooks) + mask	TBD	TBD	TBD	TBD
Single	Single-SAE	SAE features + mask	TBD	TBD	TBD	TBD
Hybrid	Loss + Entropy	Loss + entropy + mask	TBD	TBD	TBD	TBD
Hybrid	Loss + SAE	Loss + SAE + mask	TBD	TBD	TBD	TBD
Hybrid	Entropy + SAE	Entropy + SAE + mask	TBD	TBD	TBD	TBD
Hybrid	Loss + Entropy + SAE	3-stream fusion + mask	TBD	TBD	TBD	TBD

Table 1. Two-column comparison table between baselines and proposed models. All pairwise-supervised sources are converted with the same ReLU-based objective and aligned to the 1–5 target range before training/evaluation.

Concretely, we construct a reliability-style diagram [13] by binning steps according to predictive entropy H_t and reporting the mean token loss ℓ_t and mean clip-level human rating within each bin. A well-calibrated model would produce a monotone increase in ℓ_t with H_t ; deviations from monotonicity—particularly large ℓ_t in the lowest- H_t bins—localize the overconfidence regime described above (see Fig. ??, TBD). This analysis gives a principled reason why neither loss nor entropy alone suffices: each signal misreads one axis of the confidence-correctness plane, and their *disagreement* is itself an informative feature.

5.2 Temporal View: Entropy Curves as Uncertainty Signatures

Aggregate statistics collapse a sequence into a single number and therefore miss when the model is uncertain. The temporal shape of $\mathbf{h} = [H_1, \dots, H_T]$ carries additional information that correlates with human ratings beyond its mean or variance.

Recent work supports treating entropy as a time-resolved reliability signal. Entropy has been used at decoding time to suppress distracted or noisy generations [15], and at the training dynamics level, an empirical law $R = -a \exp(H) + b$ between policy entropy H and downstream performance R has been established for reasoning language models [16], indicating that entropy and output mutual information are structurally, not coincidentally, linked. In the generative setting, teacher-forced training and free-running inference produce systematically different sequence distributions [17], meaning that the teacher-forced loss curve alone can misrepresent the model’s behavior under its own rollouts. Entropy, computed directly from the predictive distribution, is comparatively less affected by this exposure-bias gap and thus a useful complementary temporal signal.

Empirically, we analyze entropy trajectories in three ways:

- 1. Rating-stratified averages.** Clips are grouped by human-rating quartile and mean entropy curves are plotted after length normalization. Higher-rated clips exhibit entropy trajectories with moderate local variation around structural boundaries; lower-rated clips tend to show either near-flat entropy (symptomatic of mode collapse or repetition) or erratic os-

cillation.

- 2. Curve statistics as features.** We extract temporal descriptors—variance, lag- k autocorrelation, peak density, and spectral centroid of the entropy curve—and ablate them against the raw curve. The lift they provide over $\{\text{mean}(H), \text{var}(H)\}$ quantifies how much *temporal structure*, not level, drives the prediction.
- 3. Loss–entropy coupling dynamics.** Using the clip-level coupling $\rho_{\ell, H}^{(i)}$ introduced earlier, we inspect how ρ distributes across rating quartiles and where within the sequence the two curves diverge the most.

Together, these diagnostics convert entropy from a scalar summary into a time-resolved uncertainty signature that reflects how the model navigates musical structure during generation.

5.3 Representational View: SAE Latents as Good-vs-Bad Probes

Loss and entropy are logit-level summaries: they compress the full predictive distribution into one number per step. SAE features instead act on the hidden state u_t and provide a *representational* readout that is not available from the output alone. This motivates a third analysis question: do the sparse latents $\{z_t^{(k)}\}$ contain directions that already separate human-preferred from human-dispreferred clips, without any supervision from human ratings during SAE training?

To answer this, we perform a latent-level correlation analysis. For each SAE unit k , let

$$\bar{z}^{(k,i)} = \frac{1}{T_i} \sum_{t=1}^{T_i} z_t^{(k,i)} \quad (15)$$

denote its mean activation on clip i , and let y_i be the clip’s human rating. We score each unit by its Pearson correlation $r_k = \text{corr}(\{\bar{z}^{(k,i)}\}_i, \{y_i\}_i)$ and rank units by $|r_k|$ to isolate the most rating-aligned directions, separately for $r_k > 0$ and $r_k < 0$.

Three probes follow:

- Top-ranked latent profiles.** For the top positive and top negative latents, we report activation distribu-

tions on clips in the highest and lowest rating quartiles. Clear bimodality indicates that the pretrained model already encodes a good-vs-bad axis that can be read out linearly.

- **Activation-ranked retrieval.** For a given top latent k^* , we retrieve the clips with the largest $\bar{z}^{(k^*, i)}$ and inspect them qualitatively. This turns an abstract latent index into an interpretable musical property (e.g., stable harmonic motion vs. abrupt timbre changes).

- **Latent-subset ablation.** We compare Single-SAE trained on the full SAE dictionary against Single-SAE restricted to the top- K rating-correlated latents. The relative gap quantifies how concentrated the human-aligned information is within the sparse dictionary.

Unlike the calibration and temporal views—which explain *when* the model is confident or surprised—the representational view asks a different question: *whether the model, through its hidden states alone, already has an internal notion of what humans will rate as good or bad music.* Our results suggest that this notion exists and is at least partially linear, which is why SAE features yield gains that are complementary to, rather than overlapping with, loss and entropy.

6. FUTURE WORK

7. ACKNOWLEDGEMENT

To be completed for the camera-ready version (omit in anonymous submission).

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